

# EFFICIENT BAYESIAN INFERENCE FROM NOISY PAIRWISE COMPARISONS

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## ABSTRACT

Human preference data are essential for evaluating generative models, because standard automatic metrics still miss decisive aspects of perceived quality. Yet human studies are costly and intrinsically noisy: participants differ in expertise, attention, and diligence. Pairwise comparisons offer a strong evaluation primitive, but converting sparse and noisy comparisons into trustworthy quality scores requires an aggregation model that treats rater variability explicitly. Existing Bradley-Terry-based methods either ignore this variability or rely on optimization procedures without convergence guarantees, leaving rankings less robust and less interpretable than modern human evaluation demands. We introduce BBQ, a Bayesian Bradley-Terry model that directly estimates rater quality, downweights or removes unreliable participants, and delivers guaranteed monotonic likelihood improvement through an Expectation-Maximization algorithm. Across human-evaluation datasets, BBQ converges faster, produces well-calibrated uncertainty estimates, and yields more robust and interpretable rankings than baseline Bradley-Terry models, including in noisy crowdsourced settings. BBQ therefore provides a principled and practical foundation for more reliable, cost-effective human evaluation of generative models.

## 1 INTRODUCTION

Generative models, especially large language models (LLMs) and image generators, require evaluation procedures that capture human preference rather than only proxy statistics. Metrics such as BLEU (Papineni et al., 2002) and perplexity (Jelinek, 1998) for LLMs, or PSNR (Gonzalez, 2009), MS-SSIM (Wang et al., 2003), and FID (Heusel et al., 2017) for image models, provide only a partial view of perceived quality (Mentzer et al., 2020; CLIC, 2025; Chiang et al., 2024). Human evaluation therefore remains the decisive instrument for establishing meaningful rankings between models. This is why leaderboards and comparative studies of large language models and Learned Image Compression (LIC) methods now place human preference data at the center of evaluation.

Human evaluations must also be efficient, because collecting high-quality judgments is expensive, slow, and vulnerable to subjective noise. Pairwise comparisons meet this need particularly well: participants choose between two items instead of assigning absolute scores, which makes the task easier and typically produces more consistent judgments (Zerman et al., 2018; Wang et al., 2023). However, exhaustive pairwise collection is not viable, since the number of pairs grows quadratically with the number of items. Under this sparse-comparison regime, simple win rates are an inadequate basis for global quality scores because each win rate is inseparable from the strength of the opponents seen by that item.

Leaderboards therefore need aggregation methods that transform sparse pairwise outcomes into stable global rankings. The Bradley-Terry (BT) (Bradley & Terry, 1952) model is the canonical solution: it estimates item scores directly from comparison outcomes and has become a standard framework for pairwise-judgment aggregation across domains.

The standard Bradley-Terry model is powerful, but its practical limitations are exactly the limitations that matter in human evaluation. BT parameters can be estimated with iterative scaling methods (Ford Jr, 1957), gradient descent, or the minorization maximization (MM) algorithm (Hunter, 2004). MM guarantees convergence to the global maximum for the basic BT model, while gradient-based

methods, closely related to the Elo rating system, remain common in practice despite lacking comparable guarantees (Hunter, 2004). For extended BT models with home field advantage (Agresti, 2010), multiple comparisons (Plackett, 1975; Luce et al., 1959), or ties (Rao & Kupper, 1967), even MM algorithms may only reach local optima (Hunter, 2004). Human evaluation introduces a second, equally important failure mode: raters may rush, guess, or lose focus, creating large differences in reliability. Ignoring this variability gives unreliable raters the same influence as careful raters, which can directly distort item scores and destabilize rankings.

We address these limitations with a Bayesian Bradley–Terry model that jointly estimates item quality and rater reliability. To the best of our knowledge, this is the first BT framework that combines an explicit Bayesian rater-quality model with closed-form EM updates. Unlike noisy-rater approaches based on gradient optimization, our method provides stable parameter estimation with guaranteed monotonic likelihood improvement and directly interpretable rater-quality parameters. The Bayesian formulation supplies regularization and principled epistemic uncertainty estimates. Our rater-specific parameters quantify how consistently and trustworthily each participant behaves, so each comparison receives an influence commensurate with the evidence that it reflects an informed judgment. The resulting latent-variable EM algorithm efficiently estimates both item and rater parameters while preserving the convergence advantages needed for reliable deployment. This design keeps the useful signal from partially reliable raters and suppresses the harm caused by inattentive or inconsistent participants, yielding rankings that are both more robust and easier to interpret across studies.

We validate the method on multiple human-evaluation datasets and introduce a newly collected *inhomogeneous rater quality (IHQ)* dataset<sup>1</sup>. The experiments show that BBQ converges faster, resists noisy comparisons more effectively, and produces more consistent rankings than standard BT and naive aggregation methods. The same framework applies beyond generative-model evaluation wherever noisy pairwise comparisons must be converted into reliable global rankings. Overall, BBQ advances human evaluation by making preference studies more cost-efficient, interpretable, and reproducible.

## 2 RELATED WORK

Ranking is a central objective in human studies, and pairwise comparison is one of the strongest protocols for obtaining these rankings. Asking participants to choose between two items rather than assign an isolated score increases sensitivity to subtle differences and reduces response variability (Zerman et al., 2018; Wang et al., 2023). This strength explains why many machine-learning leaderboards rely on pairwise preferences and then aggregate those preferences into global rankings. Chatbot Arena (Chiang et al., 2024) and the CLIC image compression challenge (CLIC, 2025) are prominent examples, both using pairwise comparisons and Bradley-Terry (BT) variants to produce model rankings.

The Bradley-Terry model (Bradley & Terry, 1952), originally developed for ranking sports competitors, remains a foundational probabilistic model for estimating whether one item is preferred over another. By converting pairwise outcomes into item scores, BT has become a core tool in games, consumer-preference analysis, and many other ranking applications. Subsequent extensions have broadened this foundation substantially. The Plackett-Luce model generalizes BT from pairwise comparisons to rankings over multiple items (Plackett, 1975; Luce et al., 1959), defining a distribution over permutations through successive BT probabilities (Luce et al., 1959). Other extensions target concrete evaluation settings, including home-field advantages (Agresti, 2010), ties (Rao & Kupper, 1967), and comparisons between groups rather than individuals (Huang et al., 2006).

Maximum likelihood estimation (MLE) is the standard route for fitting the basic BT model. Zermelo (1929) introduced the iterative estimation procedure that became widely adopted, and Lange et al. (2000) later showed that this procedure is a special case of minorization-maximization (MM), which optimizes surrogate functions to increase likelihood. Hunter (2004) extended MM algorithms to generalized BT models and identified conditions that guarantee convergence to the MLE. For the classical BT model, convexity ensures convergence to the global optimum (Hunter, 2004). For richer

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<sup>1</sup>The IHQ dataset will be released upon acceptance.

extensions involving home-field advantage, multiple comparisons, or ties, the same machinery can lose this global guarantee and converge only to a local maximum (Hunter, 2004).

Bayesian BT formulations add the regularization and uncertainty quantification that deterministic ranking pipelines often lack (Adams, 2005; Guiver & Snelson, 2009; Caron & Doucet, 2012). Caron & Doucet (2012) established that MM algorithms can be interpreted as Expectation-Maximization (EM) procedures, which enables Bayesian inference via Gibbs sampling. This view yields tractable complete-data likelihoods and convergence guarantees for the resulting Markov Chain Monte Carlo methods. The Bayesian perspective smooths posterior estimates, reduces vulnerability to unstable optima, and provides uncertainty estimates for item skills.

Annotator-quality modeling is the crucial extension for human preference studies, because the assumption that every comparison is equally reliable is rarely defensible in crowdsourced evaluation. Participant quality can vary widely, and models must account for this variation rather than averaging it away (Chen et al., 2013). Chen et al. (2013) proposed a Bayesian model that jointly estimates item quality and rater reliability with annotator-specific parameters. Their model uses Gaussian priors on item and rater parameters and likelihood scaling factors that modulate annotator influence. Posterior inference is performed with gradient-based optimization, which is sensitive to local optima in high-dimensional spaces and provides no convergence guarantee. Our EM-based formulation directly addresses this gap by providing guaranteed monotonic likelihood improvement. The Elo-based rater model from Google Research implements a simplified version of this approach and is widely used in the Challenge on Learned Image Compression (CLIC) (CLIC, 2025) and related work (Mentzer et al., 2020; Ballé et al., 2025).

BBQ fills a specific and important gap in this literature: to the best of our knowledge, no prior Bradley–Terry method combines Bayesian rater-quality modeling with closed-form EM updates. Unlike Chen et al. (2013), which uses Gaussian priors and gradient-based optimization without convergence guarantees, BBQ uses conjugate priors and an EM algorithm with guaranteed monotonic likelihood improvement. Unlike the Elo-style rater model used in CLIC (CLIC, 2025), which is a simplified heuristic update rule, BBQ supplies a principled Bayesian treatment with item uncertainty estimates and interpretable rater-quality parameters.

### 3 METHODOLOGY

#### 3.1 BRADLEY-TERRY MODEL WITH RATER QUALITY

Converting noisy pairwise comparisons into a reliable item ranking is the central statistical problem addressed by BBQ. Consider  $K$  items compared in pairs by  $R$  raters, with the observed comparison outcomes denoted by  $D$ . For two items  $i$  and  $j$ , the Bradley-Terry model (Bradley & Terry, 1952) specifies the win probability as:

$$P(i \text{ beats } j) = \frac{\lambda_i}{\lambda_i + \lambda_j} \quad (1)$$

where  $\lambda_k > 0$  denotes the skill of item  $k \in \{1, 2, \dots, K\}$ . This model gives an interpretable route from wins and losses to item scores, but its equal-reliability assumption is too strong for human evaluation. Participants differ in expertise, attention, and diligence, and these differences produce judgments with unequal reliability.

BBQ makes this reliability explicit through a rater-specific quality parameter  $q_r \in [0, 1]$ . With probability  $q_r$ , rater  $r$  contributes an informed judgment governed by the Bradley-Terry model; with probability  $1 - q_r$ , the rater behaves like a random guesser choosing uniformly between the two items. This yields the following mixture probability for rater  $r$  ranking item  $i$  above item  $j$ :

$$P(r \text{ ranks } i \text{ above } j) = q_r \left( \frac{\lambda_i}{\lambda_i + \lambda_j} \right) + (1 - q_r) \left( \frac{1}{2} \right) \quad (2)$$

The observed-data log-likelihood is therefore:

$$\begin{aligned} \log P(D \mid \lambda, q) = & \sum_{r=1}^R \sum_{i=1}^K \sum_{\substack{j=1 \\ j \neq i}}^K \left[ w_{r,ij} \log \left( q_r \frac{\lambda_i}{\lambda_i + \lambda_j} + (1 - q_r) \frac{1}{2} \right) \right. \\ & \left. + (n_{r,ij} - w_{r,ij}) \log \left( q_r \frac{\lambda_j}{\lambda_i + \lambda_j} + (1 - q_r) \frac{1}{2} \right) \right], \end{aligned} \quad (3)$$

where  $n_{r,ij}$  is the number of comparisons between items  $i$  and  $j$  made by rater  $r$ , and  $w_{r,ij}$  is the number of times rater  $r$  ranks item  $i$  above item  $j$ . This likelihood has no closed-form maximum likelihood estimator, so  $\lambda$  and  $q$  must be estimated iteratively. Gradient descent is possible, but it introduces learning-rate sensitivity and does not guarantee convergence. EM is the natural alternative: it avoids learning-rate tuning, guarantees monotonic improvement of the observed-data likelihood, and converges to a stationary point (Dempster et al., 1977).

### 3.2 THURSTONIAN INTERPRETATION

Following Caron & Doucet (2012), we use a Thurstonian interpretation of the Bradley-Terry model (Diaconis, 1988) because it exposes the latent variables needed for an efficient EM derivation. A comparison between items  $i$  and  $j$  is treated as a race in which the items have random arrival times  $Y_i$  and  $Y_j$ . These arrival times follow exponential distributions:

$$Y_i \sim \mathcal{E}(\lambda_i), \quad Y_j \sim \mathcal{E}(\lambda_j), \quad (4)$$

and the item with the smaller arrival time wins. This construction directly recovers the standard Bradley-Terry probability:

$$P(i \text{ beats } j) = P(Y_i < Y_j) = \frac{\lambda_i}{\lambda_i + \lambda_j}. \quad (5)$$

For EM, we introduce latent variables that make the complete-data likelihood tractable. First, define the indicator variable

$$A_{r,ij}^{(c)} \sim \text{Bernoulli}(q_r), \quad (6)$$

which records whether the  $c$ -th comparison of items  $i$  and  $j$  by rater  $r$  follows the Bradley-Terry component rather than random guessing. Using these indicators, we define  $Z_{r,ij}$  as the accumulated minimal arrival time across the  $n_{r,ij}$  comparisons by rater  $r$ :

$$Z_{r,ij} = \sum_{c=1}^{n_{r,ij}} A_{r,ij}^{(c)} \min(Y_i^{(c)}, Y_j^{(c)}). \quad (7)$$

Conditioned on  $m_{r,ij} = \sum_{c=1}^{n_{r,ij}} A_{r,ij}^{(c)}$ , the variable  $Z_{r,ij}$  follows a Gamma distribution,

$$Z_{r,ij} \mid m_{r,ij} \sim \Gamma(m_{r,ij}, \lambda_i + \lambda_j), \quad (8)$$

where  $\Gamma(\alpha, \beta)$  denotes the Gamma distribution with shape parameter  $\alpha$  and inverse scale  $\beta$ . This Gamma latent-variable formulation is what makes the EM update tractable while preserving explicit rater-quality modeling.

### 3.3 EXPECTATION-MAXIMIZATION UPDATES

The EM algorithm computes maximum a posteriori (MAP) estimates of  $\lambda$  and  $q$  by treating rater-quality assignments and unobserved Thurstonian arrival times as latent variables. This construction gives a principled estimator with guaranteed convergence to a stationary point of the posterior distribution.

We place conjugate priors on the parameters to obtain closed-form updates. The item skills  $\lambda_k$  receive a Gamma prior,  $\lambda_k \sim \Gamma(a, b)$ , conjugate to the exponential distribution, and each rater-quality parameter  $q_r$  receives a Beta prior,  $q_r \sim B(\alpha, \beta)$ .

The algorithm repeatedly maximizes the expected complete-data log-posterior conditioned on the current parameter estimates  $(\lambda^*, q^*)$ . It alternates between the following two steps until convergence:

**E-step: Expectation** In the E-step, we compute the expected complete-data log-posterior  $Q$ . This quantity is the posterior objective for all observed and latent variables, averaged under the current parameter estimates from the previous iteration:

$$Q(\lambda, q \mid \lambda^*, q^*) = \mathbb{E}_{A, Z \mid D, \lambda^*, q^*} [\ell_c(\lambda, q; D, Z, A) + \log P(\lambda) + \log P(q)]. \quad (9)$$

We decompose the complete-data log-likelihood  $\ell_c$  into three components.

$$\ell_c(\lambda, q; D, Z, A) = \log P(Z \mid D, A, \lambda, q) + \log P(A \mid D, \lambda, q) + \log P(D \mid \lambda, q). \quad (10)$$

**M-step: Maximization** The M-step updates the model parameters by maximizing  $Q$ . The expectations from the E-step turn the original coupled optimization problem into direct closed-form updates. Both updates depend on the posterior probability that a comparison from rater  $r$  was generated by the Bradley-Terry component. This probability, denoted  $\gamma$ , is the effective weight assigned to the rater’s judgment according to its agreement with the model’s current predictions:

$$\gamma_{r,ij}^{(t-1)} = \frac{q_r^{(t-1)} y_{ij}^{(t-1)}}{q_r^{(t-1)} y_{ij}^{(t-1)} + (1 - q_r^{(t-1)}) \frac{1}{2}}, \quad (11)$$

where  $y_{ij}^{(t-1)} = \frac{\lambda_i^{(t-1)}}{\lambda_i^{(t-1)} + \lambda_j^{(t-1)}}$  is the Bradley-Terry probability that item  $i$  beats item  $j$ . Thus  $\gamma$  quantifies the evidence that a comparison was meaningful rather than random under the current model. Higher  $\gamma$  values give trustworthy comparisons more influence.

The rater-quality update for  $q_r$  is a regularized weighted average of meaningful judgments. The numerator accumulates the rater’s effective Bradley-Terry-consistent comparisons, with each contribution weighted by  $\gamma$ , and adds the Beta-prior contribution. The denominator normalizes by the rater’s total comparisons and prior strength. Consequently, raters whose choices repeatedly align with the model receive higher quality estimates. The update is:

$$q_r^{(t)} = \frac{\sum_{i=1}^K \sum_{j=i+1}^K [w_{r,ij} \gamma_{r,ij}^{(t-1)} + w_{r,ji} \gamma_{r,ji}^{(t-1)}] + (\alpha - 1)}{n_r + \alpha + \beta - 2}, \quad (12)$$

where  $n_r$  is the total number of comparisons by rater  $r$ .

The item-skill update for  $\lambda_i$  balances effective wins against exposure. The numerator aggregates the effective wins of item  $i$  across raters, again weighted by  $\gamma$ , and includes the Gamma-prior contribution. The denominator accounts for the comparisons involving item  $i$  and provides the corresponding normalizing term. This yields the closed-form update:

$$\lambda_i^{(t)} = \frac{\sum_{r=1}^R \left[ \sum_{j=1, j \neq i}^K w_{r,ij} \gamma_{r,ij}^{(t-1)} \right] + (a - 1)}{\sum_{j=1, j \neq i}^K \left[ \frac{w_{r,ij} \gamma_{r,ij}^{(t-1)} + w_{r,ji} \gamma_{r,ji}^{(t-1)}}{\lambda_i^{(t-1)} + \lambda_j^{(t-1)}} \right] + b}. \quad (13)$$

Appendix B gives the full Expectation Maximization derivation.

## 4 EXPERIMENTAL RESULTS

We evaluate **Bayesian Bradley-Terry** with rater **Quality (BBQ)** against two strong BT-based baselines: Bayesian Bradley-Terry (Bayes-BT) (Caron & Doucet, 2012) and Crowd-BT, a gradient-descent BT model with rater quality (Chen et al., 2013). For Crowd-BT, we use the Google Research implementation (Google, 2025) adopted by both CLIC (2025) and Ballé et al. (2025). Section C reports the hyperparameters used throughout the experiments.

The evaluation spans human-preference benchmarks for large language models (HUMAINE (Team, 2025), MT-Bench (Zheng et al., 2023)) and image compression (WD (Ballé et al., 2025), HiFiC (Mentzer et al., 2020), ConHa (Aczel & Wattenhofer, 2024)), together with our newly collected inhomogeneous rater quality (IHQ) dataset based on the CLIC2024 image test set (CLIC, 2025). We split IHQ into screened and unscreened subsets to isolate the effect of rater quality. This design directly tests performance under high-quality, low-quality, and mixed-comparison regimes. Section E

describes the IHQ collection protocol, and Section D together with Table 2 summarizes all datasets, comparisons, raters, and items.

Because real datasets do not provide ground-truth rankings, we first verify the models on simulations, where all methods recover the same ordering as sample size increases. For crowdsourced datasets, we use the ranking from the full dataset as the reference ranking. The top-1 item provides a concrete validation of this reference: for every dataset except IHQ-unscreened, all three models recover the same top-ranked item. On IHQ-unscreened, the distinction is decisive: Crowd-BT and Bayes-BT fail to identify the reference image as the highest-quality item, while BBQ identifies it correctly.

A reliable aggregation method should reproduce the same ordering when the study is repeated. We measure this stability with bootstrapping, which estimates ranking variability under repeated sampling. Section F gives the bootstrapping procedure. The WD and HiFiC datasets require cautious interpretation because their original studies used active pair selection: HiFiC used binary search, and WD selected pairs by maximum information gain.

We report two complementary stability metrics. Top-1 agreement is the fraction of bootstrap samples that select the same best item as the full dataset. Kendall’s Tau ( $\tau$ ) measures ordinal correlation between rankings (Kendall, 1938), with larger values indicating more stable rankings. Top-1 agreement captures reliability of the winner, while Kendall’s Tau captures stability of the full ordering.

#### 4.1 PERFORMANCE ACROSS DATASETS

Table 1 compares ranking accuracy and stability across datasets using Top-1 agreement and Kendall’s Tau. Datasets with more comparisons per model, such as MT-Bench, WD, and HiFiC, are generally easier for all methods. HUMAINE breaks this simple pattern, showing that comparison volume alone does not determine performance; rater consistency and diversity also materially affect aggregation quality.

Overall, BBQ is the most stable and robust method across datasets. It identifies the top-performing item most often, including shared-best performance on three datasets. It also achieves the best overall ranking stability on five of eight datasets and ranks second on the remaining three.

The three datasets where BBQ ranks second in Kendall’s Tau are precisely the datasets with high-quality, homogeneous rater pools. On MT-Bench, all methods perfectly recover the top item. The WD dataset, collected by Ballé et al. (2025), contains only five raters, likely the paper’s authors, which indicates exceptionally careful evaluation. IHQ-screened explicitly filters raters for quality. These cases confirm the intended behavior of BBQ: when rater quality is already uniformly high, explicit rater modeling is less necessary and simpler models can close the gap.

The IHQ experiments make the value of BBQ especially clear. When low-quality raters are present, as in CLIC-all and CLIC-unscreened, BBQ substantially outperforms both Crowd-BT and Bayes-BT. All methods improve on the screened subset, confirming that rater selection matters. However, Crowd-BT still degrades on the full dataset despite modeling rater quality, likely because crowd-sourced raters provide fewer than 40 comparisons each and therefore give gradient-based estimation too little reliable per-rater evidence. BBQ remains strong without screening, with only a small Top-1 decrease, demonstrating that its Bayesian EM formulation is robust to low-quality raters.

Uniformly high rater quality is difficult to guarantee in practice. Large-scale crowdsourcing naturally introduces variability, screening increases cost, and subjective factors can affect even diligent annotators. BBQ directly addresses this practical regime by preserving useful information from partially reliable raters while suppressing noisy contributions.

#### 4.2 SCALING WITH RATERS AND COMPARISONS

Table 1 shows that more comparisons per model generally improve performance. Additional evidence can come from two sources: adding raters or asking each rater for more comparisons.

Figure 1 separates these two sources of evidence for the three BT variants. The left column varies the number of raters while fixing comparisons per rater at the maximum, and the right column varies comparisons per rater while fixing the number of raters at the maximum. Top-1 accuracy clearly

Table 1: Performance of BT-based aggregation methods across datasets. Top: Top-1 agreement [%], Bottom: Kendall’s Tau. BBQ most often identifies the top-performing item, achieves the best overall ranking stability on more than half of the datasets, and remains the runner-up on the rest.

|                     | HUMAINE       | MT-Bench      | WD            | HiFiC         | ConHa         | IHQ           |               |               |
|---------------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|
|                     |               |               |               |               |               | all           | scr.          | unscr.        |
| Top-1 Agreement [%] |               |               |               |               |               |               |               |               |
| Crowd-BT            | 85.60         | <b>100.00</b> | 99.29         | 98.63         | 66.59         | 85.31         | 98.57         | 33.15         |
| Bayes-BT            | <u>97.30</u>  | <b>100.00</b> | <b>100.00</b> | <b>100.00</b> | 57.30         | 75.07         | <u>98.90</u>  | 24.32         |
| BBQ (ours)          | <b>97.50</b>  | <b>100.00</b> | <b>100.00</b> | <b>100.00</b> | <b>77.12</b>  | <b>99.32</b>  | <b>99.80</b>  | <b>61.92</b>  |
| Kendall’s Tau       |               |               |               |               |               |               |               |               |
| Crowd-BT            | 0.9385        | <b>0.9743</b> | 0.9279        | <u>0.9366</u> | 0.9180        | <u>0.9245</u> | <b>0.9171</b> | 0.8482        |
| Bayes-BT            | <u>0.9443</u> | 0.9569        | <b>0.9459</b> | <u>0.9293</u> | 0.9110        | <u>0.9240</u> | 0.9116        | <u>0.8507</u> |
| BBQ (ours)          | <b>0.9463</b> | <u>0.9675</u> | <u>0.9359</u> | <b>0.9525</b> | <b>0.9265</b> | <b>0.9270</b> | <u>0.9132</u> | <b>0.8563</b> |

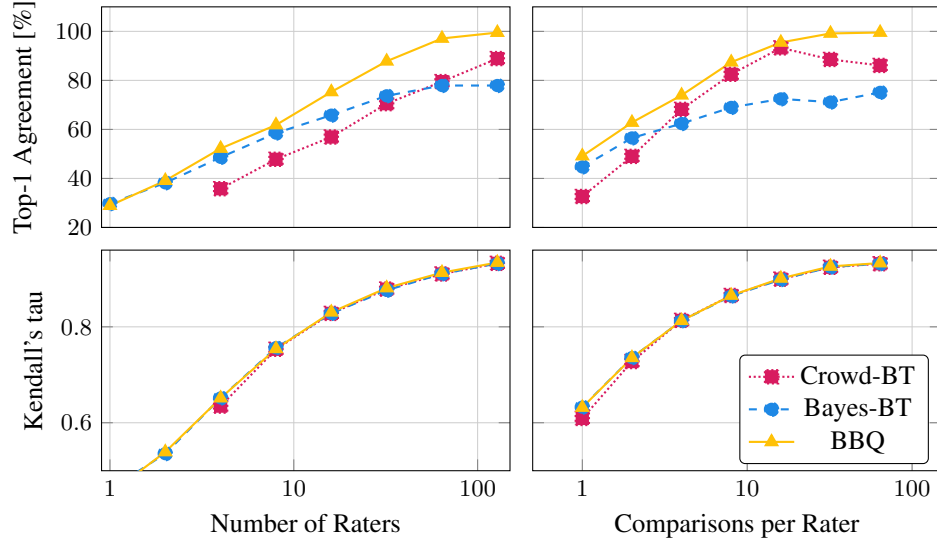


Figure 1: Scaling behavior of Bradley–Terry variants (Crowd-BT (Caron & Doucet, 2012), Bayes-BT (Chen et al., 2013), BBQ (ours)) on IHQ-all. **Left:** Performance as the number of raters increases. **Right:** Performance as comparisons per rater increase. More raters and more comparisons both improve Top-1 agreement and Kendall’s  $\tau$ . Top-1 agreement separates the models most clearly, while Kendall’s  $\tau$  remains closer across methods. Crowd-BT fails to converge with very few raters, underscoring the practical advantage of EM. Under sparse data Crowd-BT and BBQ are similar, but BBQ overtakes Bayes-BT as rater or comparison counts grow.

differentiates the methods, whereas Kendall’s  $\tau$  remains comparatively similar. Crowd-BT fails to converge with only one or two raters, directly illustrating the advantage of an EM algorithm with convergence guarantees over gradient descent.

Crowd-BT and BBQ are similar in the sparsest regimes, where the data contain too little per-rater evidence to reliably distinguish rater qualities. BBQ benefits once there are both multiple raters and multiple comparisons per rater, and its advantage over Bayes-BT grows as either source of evidence increases. Bayes-BT generally underperforms, but it can surpass Crowd-BT when few raters or limited per-rater comparisons make explicit rater-quality estimation unreliable for gradient-based Crowd-BT.

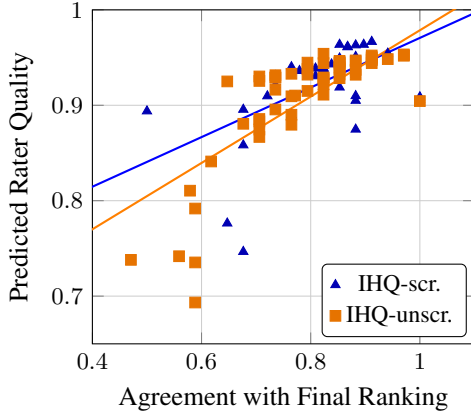


Figure 2: Rater agreement with the final ranking (x-axis) versus predicted rater quality (y-axis) on the IHQ datasets. Each point is one rater. Triangles denote the filtered dataset, and squares denote the unfiltered dataset, showing that BBQ assigns lower quality to less consensus-aligned raters.

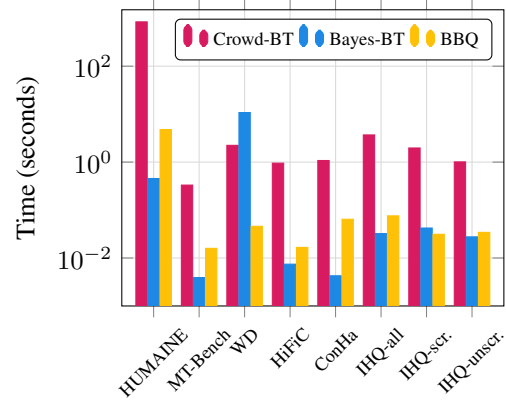


Figure 3: Average computation time in seconds (log-scale) for the three models on one bootstrapped sample across eight datasets. Crowd-BT can be computationally expensive on large datasets, whereas BBQ remains consistently fast across all datasets.

#### 4.3 RATER QUALITY

Figure 2 confirms that the learned rater-quality parameter has the intended interpretation. Raters who agree more strongly with the consensus ranking receive higher predicted quality. The filtered dataset yields a Pearson correlation of  $r = 0.551$  across 50 raters, and the unfiltered dataset yields an even stronger correlation of  $r = 0.724$  across 62 raters.

The unscreened dataset contains several raters with both low agreement and low predicted quality, exactly the failure mode that motivates BBQ. Some IHQ-unscreened raters fall below random-guessing agreement (50%), indicating systematic disagreement with the majority. BBQ identifies these raters and assigns them lower quality scores, reducing their influence on the final ranking and directly mitigating the noise they introduce.

#### 4.4 COMPUTATIONAL EFFICIENCY

Figure 3 reports the average time each method needs to process one bootstrapped sample across datasets. These measurements show that BBQ is practical for real evaluation workflows. BBQ converges within a few seconds on every dataset. Crowd-BT is slower across the board and becomes especially costly on comparison-heavy datasets such as HUMAINE, where convergence takes around 15 minutes. Bayes-BT is slow on WD and slightly faster than BBQ on some other datasets. BBQ’s efficiency follows directly from its closed-form EM updates, which deliver rapid convergence even on large datasets.

This efficiency is especially notable because Crowd-BT is highly optimized and implemented in C (Kernighan & Ritchie, 1988), while BBQ uses plain NumPy without targeted optimization. A compiled or more aggressively vectorized BBQ implementation should therefore be even faster. Together with its robustness and stability, this speed makes BBQ highly practical for large-scale human preference studies and repeated-bootstrap workflows. The combination of accuracy, stability, interpretability, and speed makes BBQ a strong choice for real-world deployment.

#### 4.5 CONFIDENCE INTERVALS

We also test whether each method provides usable uncertainty estimates for item rankings. In a controlled simulation, raters compare equally skilled items under  $H_0$ . Each model estimates skills and constructs uncertainty intervals for testing significant differences (Section G). Figure 6 reports

the resulting Type I error rates. At the 99% confidence level, the expected error rate is 1%. Crowd-BT and BBQ track this target closely with slightly lower rates, while Bayes-BT is substantially more conservative at  $\sim 0.1\%$ . Thus, BBQ matches Crowd-BT in producing well-calibrated uncertainty estimates, whereas Bayes-BT is overly conservative for significance testing.

## 5 LIMITATIONS

BBQ is designed for the realistic case where rater quality varies, so its advantage naturally shrinks when all raters are uniformly reliable. In homogeneous high-quality datasets, explicit variability modeling adds little beyond simpler BT models. However, achieving such uniformly reliable rater pools typically requires substantial screening cost and effort, which limits scalability. This trade-off clarifies the method’s strongest use case: BBQ is most valuable in realistic crowdsourced evaluations and less critical in small, carefully curated studies.

A second limitation is input type. BBQ currently targets pairwise comparisons, while many human-evaluation protocols collect ratings, rankings, or multi-way judgments. Extending the same rater-quality principle to these feedback formats would broaden the model’s applicability.

Methodologically, our empirical stability analysis relies on a finite number of bootstrap resamples. This provides a practical and informative stability estimate, but larger-scale studies would further stress-test robustness. The model also assumes independent comparisons and does not explicitly represent contextual or order effects, both of which can influence human judgments in practice.

Finally, BBQ scales well in our experiments, but extremely large item or rater populations may still require optimized or compiled implementations.

## 6 CONCLUSION

We introduced BBQ, a Bayesian Bradley-Terry model that jointly estimates item quality and rater reliability. The central contribution is a closed-form Expectation-Maximization (EM) algorithm that simultaneously estimates item skills and individual rater quality, thereby down-weighting or removing the influence of unreliable participants. This explicit rater-quality model produces more stable and accurate rankings in the settings where human evaluation is most vulnerable to noise. The EM algorithm provides rapid convergence and guaranteed monotonic likelihood improvement, directly addressing the instability of gradient-based approaches.

Across diverse datasets, BBQ achieves high Top-1 agreement and the strongest overall Kendall’s Tau profile, demonstrating robust and reliable aggregation. Its advantage is clearest with noisy or crowdsourced raters, where it maintains accuracy even when individual raters provide few comparisons. BBQ is therefore especially well matched to large-scale evaluations with non-expert raters or heterogeneous attention and diligence. The results show that modeling rater quality is not a cosmetic extension; it is a central requirement when evaluation quality is heterogeneous or partially unreliable. The Bayesian formulation also provides principled uncertainty estimates for item scores, enabling interpretable comparisons across studies. Our calibration experiments show that these error bars support significance assessment, and the learned rater-quality scores align with agreement to final rankings, validating their role as interpretable reliability estimates.

Overall, BBQ advances human evaluation methodology by providing a practical, interpretable, and generalizable solution for aggregating noisy pairwise comparisons. It establishes a stronger foundation for cost-effective, reproducible, and trustworthy human studies of AI-generated content.

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## A USAGE OF LLMs

During preparation of this paper, we used large language models (LLMs) strictly as auxiliary tools. ChatGPT, Claude, Gemini, and Grammarly supported spellchecking, phrasing refinement, and text condensation for clarity and readability. ChatGPT, Claude, and Cursor also assisted with code analysis, code completion, and visualization generation during development. All core research ideas, experimental design choices, and interpretations of results were developed independently by the authors.

## B BAYESIAN BT WITH RATER QUALITY DERIVATION

We consider  $R$  raters comparing  $K$  items. Each rater  $r$  has quality  $q_r$ : with probability  $q_r$  the rater follows the Bradley–Terry model, and with probability  $1 - q_r$  the rater chooses randomly. The following EM (Dempster et al., 1977) derivation makes these rater-specific reliabilities explicit inside the Bayesian estimation framework.

### B.1 MODEL DEFINITION

We use the following notation throughout the derivation:

- $\lambda = (\lambda_1, \dots, \lambda_K)$ : skill parameters for the  $K$  items.
- $q = (q_1, \dots, q_R)$ : quality parameters for the  $R$  raters.
- $Y_i \sim \mathcal{E}(\lambda_i)$ : latent arrival time associated with item  $i$ .
- $A_{r,ij}^{(c)} \sim \text{Bernoulli}(q_r)$ : indicator that the  $c$ -th comparison of pair  $(i, j)$  by rater  $r$  follows the Bradley–Terry model.
- $n_{r,ij}$ : total number of comparisons of  $(i, j)$  by rater  $r$ .
- $n_{ij} = \sum_{r=1}^R n_{r,ij}$ : total number of comparisons of  $(i, j)$ .
- $w_{r,ij}$ : number of times rater  $r$  ranked  $i$  above  $j$ .
- $w_{ij} = \sum_{r=1}^R w_{r,ij}$ : total number of times  $i$  was ranked above  $j$ .
- $m_{r,ij} = \sum_{c=1}^{n_{r,ij}} A_{r,ij}^{(c)}$ : number of Bradley–Terry-model comparisons of  $(i, j)$  by rater  $r$ .
- $m_{ij} = \sum_{r=1}^R m_{r,ij}$ : total number of Bradley–Terry-model comparisons of  $(i, j)$ .
- $v_{r,ij} = \sum_{c=1}^{w_{r,ij}} A_{r,ij}^{(c)}$ : number of Bradley–Terry-model wins of  $i$  over  $j$  by rater  $r$ .
- $v_{ij} = \sum_{r=1}^R v_{r,ij}$ : total number of Bradley–Terry-model wins of  $i$  over  $j$ .
- $Z_{r,ij} = \sum_{c=1}^{n_{r,ij}} A_{r,ij}^{(c)} \min(Y_i^{(c)}, Y_j^{(c)})$ : sum of minimal arrival times, with

$$Z_{r,ij} \mid m_{r,ij} \sim \Gamma(m_{r,ij}, \lambda_i + \lambda_j).$$

The Bradley–Terry component assigns the following win probability:

$$P(i \text{ beats } j) = \frac{\lambda_i}{\lambda_i + \lambda_j}.$$

The rater-quality mixture then combines informed BT judgments with random choices:

$$P(r \text{ ranks } i \text{ above } j) = q_r \frac{\lambda_i}{\lambda_i + \lambda_j} + (1 - q_r) \frac{1}{2}.$$

The latent exponential representation gives the equivalent Thurstonian view:

$$Y_i \sim \mathcal{E}(\lambda_i), \quad Y_j \sim \mathcal{E}(\lambda_j), \quad P(i \text{ beats } j) = P(Y_i < Y_j).$$

## B.2 COMPLETE-DATA LOG-LIKELIHOOD

The complete-data likelihood decomposes as:

$$\begin{aligned}\ell_c(\lambda, q; D, Z, A) &= \log P(D, Z, A \mid \lambda, q) \\ &= \log P(Z \mid D, A, \lambda, q) + \log P(A \mid D, \lambda, q) + \log P(D \mid \lambda, q)\end{aligned}$$

The latent-arrival contribution is:

$$\begin{aligned}\log P(Z \mid D, A, \lambda, q) &= \sum_{i=1}^K \sum_{j=i+1}^K \left[ m_{ij} \log(\lambda_i + \lambda_j) - (\lambda_i + \lambda_j) z_{ij} \right. \\ &\quad \left. + (m_{ij} - 1) \log z_{ij} - \log \Gamma(m_{ij}) \right]\end{aligned}$$

The rater-quality assignment contribution is:

$$\log P(A \mid D, \lambda, q) = \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K \left[ m_{r,ij} \log q_r + (n_{r,ij} - m_{r,ij}) \log(1 - q_r) \right]$$

The observed-comparison contribution is:

$$\begin{aligned}\log P(D \mid \lambda, q) &= \sum_{r=1}^R \sum_{i=1}^K \sum_{\substack{j=1 \\ j \neq i}}^K \left[ v_{r,ij} \log \frac{\lambda_i}{\lambda_i + \lambda_j} + (w_{r,ij} - v_{r,ij}) \log \frac{1}{2} \right] \\ &= \sum_{r=1}^R \sum_{i=1}^K [v_{ir} \log \lambda_i] \\ &\quad - \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K [(v_{r,ij} + v_{r,ji}) \log(\lambda_i + \lambda_j) + (w_{r,ij} + w_{r,ji} - v_{r,ij} - v_{r,ji}) \log 2] \\ &= \sum_{i=1}^K [v_i \log \lambda_i] \\ &\quad - \sum_{i=1}^K \sum_{j=i+1}^K [m_{ij} \log(\lambda_i + \lambda_j) + (n_{ij} - m_{ij}) \log 2]\end{aligned}$$

Combining these terms yields the complete-data log-likelihood:

$$\begin{aligned}
\ell_c(\lambda, q; D, Z, A) &= \log P(Z \mid D, A, \lambda, q) + \log P(A \mid D, \lambda, q) + \log P(D \mid \lambda, q) \\
&= \sum_{i=1}^K \sum_{j=i+1}^K \left[ m_{ij} \log(\lambda_i + \lambda_j) - (\lambda_i + \lambda_j) z_{ij} + (m_{ij} - 1) \log z_{ij} - \log \Gamma(m_{ij}) \right] \\
&\quad + \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K \left[ m_{r,ij} \log q_r + (n_{r,ij} - m_{r,ij}) \log(1 - q_r) \right] \\
&\quad + \sum_{i=1}^K [v_i \log \lambda_i] \\
&\quad - \sum_{i=1}^K \sum_{j=i+1}^K [m_{ij} \log(\lambda_i + \lambda_j) + (n_{ij} - m_{ij}) \log 2] \\
&= \sum_{i=1}^K \sum_{j=i+1}^K \left[ (n_{ij} - m_{ij}) \log 2 - (\lambda_i + \lambda_j) z_{ij} + (m_{ij} - 1) \log z_{ij} - \log \Gamma(m_{ij}) \right] \\
&\quad + \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K \left[ m_{r,ij} \log q_r + (n_{r,ij} - m_{r,ij}) \log(1 - q_r) \right] \\
&\quad + \sum_{i=1}^K [v_i \log \lambda_i]
\end{aligned}$$

### B.3 EXPECTATION STEP

We introduce conjugate priors to keep the posterior updates closed form: Gamma distribution  $\lambda_i \sim \Gamma(a, b)$  for each item  $i$ , and Beta  $q_r \sim B(\alpha, \beta)$  for each rater  $r$ .

The  $Q$ -function is the expected complete-data log-posterior:

$$Q(\lambda, q \mid \lambda^*, q^*) = \mathbb{E}_{A, Z \mid D, \lambda^*, q^*} [\ell_c(\lambda, q; D, Z, A) + \log P(\lambda) + \log P(q)].$$

The posterior probability that an annotation is Bradley-Terry-consistent is:

$$\begin{aligned}
P(A_{r,ij}^{(k)} = 1 \mid i \succ j) &= \frac{P(i \succ j \mid A = 1) P(A = 1)}{P(i \succ j \mid A = 1) P(A = 1) + P(i \succ j \mid A = 0) P(A = 0)} \\
&= \frac{\left( \frac{\lambda_i^*}{\lambda_i^* + \lambda_j^*} \right) q_r^*}{\left( \frac{\lambda_i^*}{\lambda_i^* + \lambda_j^*} \right) q_r^* + \frac{1}{2} (1 - q_r^*)} = \gamma_{r,ij}^*.
\end{aligned}$$

The corresponding expected sufficient statistics are:

$$\begin{aligned}
\mathbb{E}[m_{r,ij}] &= w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^* \\
\mathbb{E}[m_{ij}] &= \sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) \\
\mathbb{E}[v_{r,ij}] &= w_{r,ij} \gamma_{r,ij}^* \\
\mathbb{E}[v_{ir}] &= \sum_{j, j \neq i}^K w_{r,ij} \gamma_{r,ij}^* \\
\mathbb{E}[v_i] &= \sum_{r=1}^R \sum_{j, j \neq i}^K w_{r,ij} \gamma_{r,ij}^* \\
\mathbb{E}[z_{ij}] &= \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{\lambda_i^* + \lambda_j^*}.
\end{aligned}$$

Substituting these expectations gives the expected complete-data log-posterior:

$$\begin{aligned}\mathbb{E}[\ell_c] = & \sum_{i=1}^K \sum_{j=i+1}^K \left[ -(\lambda_i + \lambda_j) \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{\lambda_i^* + \lambda_j^*} \right] \\ & + \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K \left[ (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) \log q_r \right. \\ & \quad \left. + (n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)) \log(1 - q_r) \right] \\ & + \sum_{i=1}^K \sum_{\substack{j=1 \\ j \neq i}}^K w_{r,ij} \gamma_{r,ij}^* \log \lambda_i \\ & + \text{const.}\end{aligned}$$

The priors contribute the regularization terms:

$$\begin{aligned}\mathbb{E}[\log P(\lambda)] &= \sum_{i=1}^K [(a-1) \log \lambda_i - b \lambda_i], \\ \mathbb{E}[\log P(q)] &= \sum_{r=1}^R [(\alpha-1) \log q_r + (\beta-1) \log(1 - q_r)].\end{aligned}$$

The final  $Q$ -function is therefore:

$$\begin{aligned}Q(\lambda, q \mid \lambda^*, q^*) = & \sum_{i=1}^K \sum_{j=i+1}^K \left[ -(\lambda_i + \lambda_j) \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{\lambda_i^* + \lambda_j^*} \right] \\ & + \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K \left[ (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) \log q_r \right. \\ & \quad \left. + (n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)) \log(1 - q_r) \right] \\ & + \sum_{i=1}^K \sum_{\substack{j=1 \\ j \neq i}}^K w_{r,ij} \gamma_{r,ij}^* \log \lambda_i \\ & + \sum_{i=1}^K [(a-1) \log \lambda_i - b \lambda_i] \\ & + \sum_{r=1}^R [(\alpha-1) \log q_r + (\beta-1) \log(1 - q_r)] \\ & + \text{const.}\end{aligned}$$

where

$$\gamma_{r,ij}^* = \frac{q_r^* \frac{\lambda_i^*}{\lambda_i^* + \lambda_j^*}}{q_r^* \frac{\lambda_i^*}{\lambda_i^* + \lambda_j^*} + (1 - q_r^*) \frac{1}{2}}.$$

#### B.4 M-STEP

The M-step maximizes the  $Q$ -function with respect to  $(\lambda, q)$  while holding the E-step expectations fixed.

**Update for  $q_r$**  The rater-quality update follows by maximizing  $Q$  with respect to  $q_r$  while retaining the Beta-prior contribution.

$$\begin{aligned}
Q(q \mid \lambda^*, q^*) &= \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K \left[ (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) \log q_r \right. \\
&\quad \left. + (n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)) \log(1 - q_r) \right] \\
&\quad + (\alpha - 1) \log q_r + (\beta - 1) \log(1 - q_r) + \text{const.} \\
\frac{\partial Q}{\partial q_r} &= \sum_{i=1}^K \sum_{j=i+1}^K \left[ \frac{w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*}{q_r} - \frac{n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{1 - q_r} \right] \\
&\quad + \frac{\alpha - 1}{q_r} - \frac{\beta - 1}{1 - q_r} \\
&\stackrel{0}{=} \frac{\sum_{i=1}^K \sum_{j=i+1}^K (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) + (\alpha - 1)}{q_r} \\
&\quad - \frac{\sum_{i=1}^K \sum_{j=i+1}^K (n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)) + (\beta - 1)}{1 - q_r} \\
&\Rightarrow (1 - q_r) \left( \sum_{i=1}^K \sum_{j=i+1}^K (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) + (\alpha - 1) \right) \\
&\Rightarrow q_r \left( \sum_{i=1}^K \sum_{j=i+1}^K (n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)) + (\beta - 1) \right) \\
&\Rightarrow q_r = \frac{\sum_{i=1}^K \sum_{j=i+1}^K (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) + (\alpha - 1)}{\sum_{i=1}^K \sum_{j=i+1}^K (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^* + n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)) + (\beta + \alpha - 2)} \\
&\Rightarrow q_r^{(t)} = \frac{\sum_{i=1}^K \sum_{j=i+1}^K (w_{r,ij} \gamma_{r,ij}^{(t-1)} + w_{r,ji} \gamma_{r,ji}^{(t-1)}) + (\alpha - 1)}{n_r + \beta + \alpha - 2}
\end{aligned}$$

where the last line gives the explicit update at iteration  $t$ .

**Update for  $\lambda_i$**  The item-skill update follows by maximizing  $Q$  with respect to  $\lambda_i$  while retaining the Gamma-prior contribution.

$$\begin{aligned}
Q(\lambda \mid \lambda^*, q^*) &= \sum_{i=1}^K \sum_{j=i+1}^K -(\lambda_i + \lambda_j) \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{\lambda_i^* + \lambda_j^*} \\
&\quad + \sum_{r=1}^R \sum_{i=1}^K \sum_{\substack{j=1 \\ j \neq i}}^K w_{r,ij} \gamma_{r,ij}^* \log \lambda_i \\
&\quad + \sum_{i=1}^K [(a - 1) \log \lambda_i - b \lambda_i] + \text{const.}
\end{aligned}$$

$$\begin{aligned}
\frac{\partial Q}{\partial \lambda_i} &= - \sum_{j \neq i}^K \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{\lambda_i^* + \lambda_j^*} + \sum_{r=1}^R \frac{\sum_{j \neq i}^K w_{r,ij} \gamma_{r,ij}^*}{\lambda_i} + \frac{a-1}{\lambda_i} - b \\
0 &\stackrel{!}{=} \frac{\sum_{r=1}^R \sum_{j \neq i}^K w_{r,ij} \gamma_{r,ij}^* + (a-1)}{\lambda_i} - \sum_{j \neq i}^K \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{\lambda_i^* + \lambda_j^*} - b \\
\Rightarrow \lambda_i^{(t)} &= \frac{\sum_{r=1}^R \sum_{j \neq i}^K w_{r,ij} \gamma_{r,ij}^{(t-1)} + (a-1)}{\sum_{j \neq i}^K \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^{(t-1)} + w_{r,ji} \gamma_{r,ji}^{(t-1)})}{\lambda_i^{(t-1)} + \lambda_j^{(t-1)}} + b} \\
\gamma_{r,ij}^{(t-1)} &= \frac{q_r^{(t-1)} \frac{\lambda_i^{(t-1)}}{\lambda_i^{(t-1)} + \lambda_j^{(t-1)}}}{q_r^{(t-1)} \frac{\lambda_i^{(t-1)}}{\lambda_i^{(t-1)} + \lambda_j^{(t-1)}} + (1 - q_r^{(t-1)}) \frac{1}{2}}
\end{aligned}$$

where the last line gives the explicit update at iteration  $t$ .

## C HYPERPARAMETERS

For Crowd-BT, we use the Google (2025) implementation with its default hyperparameters, preserving the established baseline configuration.

For Bayes-BT and BBQ, the relevant hyperparameters are the prior distribution parameters and the stopping thresholds. We declare both Bayes-BT and BBQ converged when no ELO score changes by more than 1 between consecutive iterations. The ELO score is computed from the skill parameter  $\lambda$  as:

$$\text{ELO} = \log(\text{skill}) \cdot \text{ELO\_SCALE\_FACTOR}. \quad (14)$$

where we set the ELO\_SCALE\_FACTOR to 400.

We use a gamma prior with shape 5 and rate 0.1 for the skill parameters in both Bayes-BT and BBQ. For BBQ, the beta prior on rater quality uses  $\alpha = 10$  and  $\beta = 2$ , encoding a light preference for reliable raters while still allowing the data to identify low-quality participants.

## D DATASETS

We evaluate the models on a diverse set of human-preference datasets covering both language and image domains. Table 2 summarizes the scale and structure of these datasets.

For natural language, we use HUMAINE (Team, 2025), a large-scale benchmark with over 100k comparisons, and MT-Bench (Zheng et al., 2023), which provides crowd-worker model comparisons on multi-turn dialogues.

For image compression, we use WD (Ballé et al., 2025), a dense expert-labeled dataset with thousands of comparisons per rater; HiFiC (Mentzer et al., 2020), which evaluates learned image codecs; and ConHa (Aczel & Wattenhofer, 2024), which focuses on conditional generative models. These datasets vary substantially in scale, rater count, and rater expertise, providing a broad testbed for robustness.

Finally, we introduce the inhomogeneous rater quality (**IHQ**) dataset, collected through a user study on CLIC2024 (CLIC, 2025) images via the *Mabyduck* platform. IHQ contains two-alternative-forced-choice (2AFC) judgments across 28 generative image-compression models. To directly measure the impact of rater quality, we provide two subsets: (i) **screened**, where raters passed pre-screening checks for attention and display quality, and (ii) **unscreened**, where all raters are included. The screened subset represents a higher-quality evaluation regime, whereas the unscreened subset captures the noisy conditions typical of large-scale human evaluations.

Table 2: Summary of datasets used in our experiments. We collected IHQ on the Mabyduck platform; it includes both 2AFC and 3AFC settings. For IHQ, screened subsets retain only raters who passed attention and display-quality pre-screening, while unscreened subsets retain all raters and preserve the noisier evaluation regime.

| dataset                           | # comparisons | # raters | # models | comp/rater | comp/model |
|-----------------------------------|---------------|----------|----------|------------|------------|
| HUMAINE (Team, 2025)              | 105,220       | 1,977    | 27       | 53.2       | 3897.0     |
| MT-Bench (Zheng et al., 2023)     | 3,355         | 65       | 6        | 51.6       | 559.2      |
| WD (Ballé et al., 2025)           | 16,659        | 5        | 30       | 3331.8     | 555.3      |
| HiFiC (Mentzer et al., 2020)      | 5,220         | 20       | 9        | 261.0      | 580.0      |
| ConHa (Aczel & Wattenhofer, 2024) | 1,531         | 40       | 8        | 38.3       | 191.4      |
| IHQ-all                           | 4,074         | 112      | 28       | 36.4       | 145.5      |
| IHQ-screened                      | 2,012         | 50       | 28       | 40.2       | 71.9       |
| IHQ-unscreened                    | 2,062         | 62       | 28       | 33.3       | 73.6       |

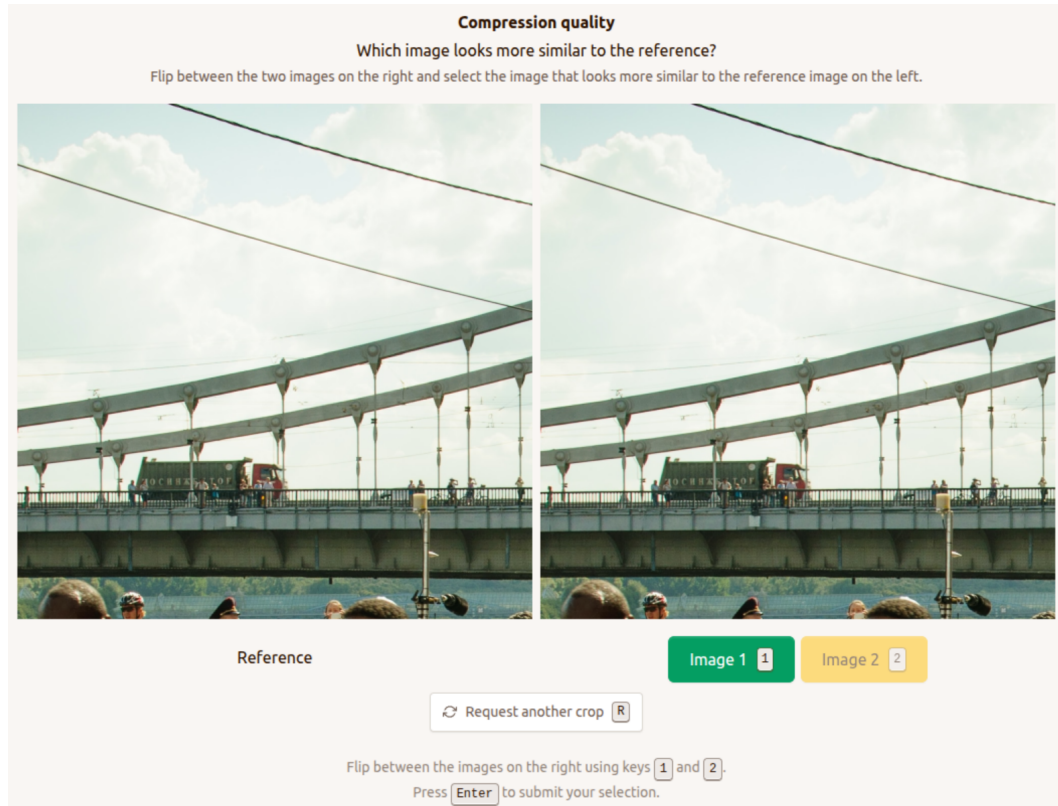


Figure 4: Mabyduck user-study interface for pairwise comparisons. The reference image appears on the left, and the rater selects which of two compressed images on the right is more similar to the reference.

This dataset collection lets us evaluate both relatively clean large-scale benchmarks and noisier regimes where rater-reliability modeling is essential.

## E USER STUDY PLATFORM

We collected all CLIC2024 (CLIC, 2025) pairwise comparisons with the Mabyduck platform (Ltd., 2025). Figure 4 shows the interface. Raters answered the question “Which image looks more similar to the reference image?”; ties were disallowed, and pairs were sampled uniformly at random.

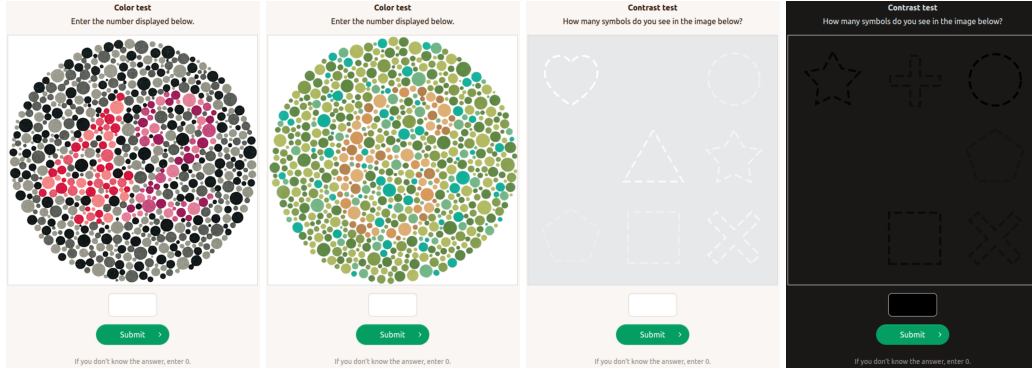


Figure 5: Four examples from the rater pre-screening process. The first two images test color vision by asking raters to identify the number in each pattern. The last two images test sensitivity to low-contrast objects: a light gray shape on white and a dark gray shape on black.

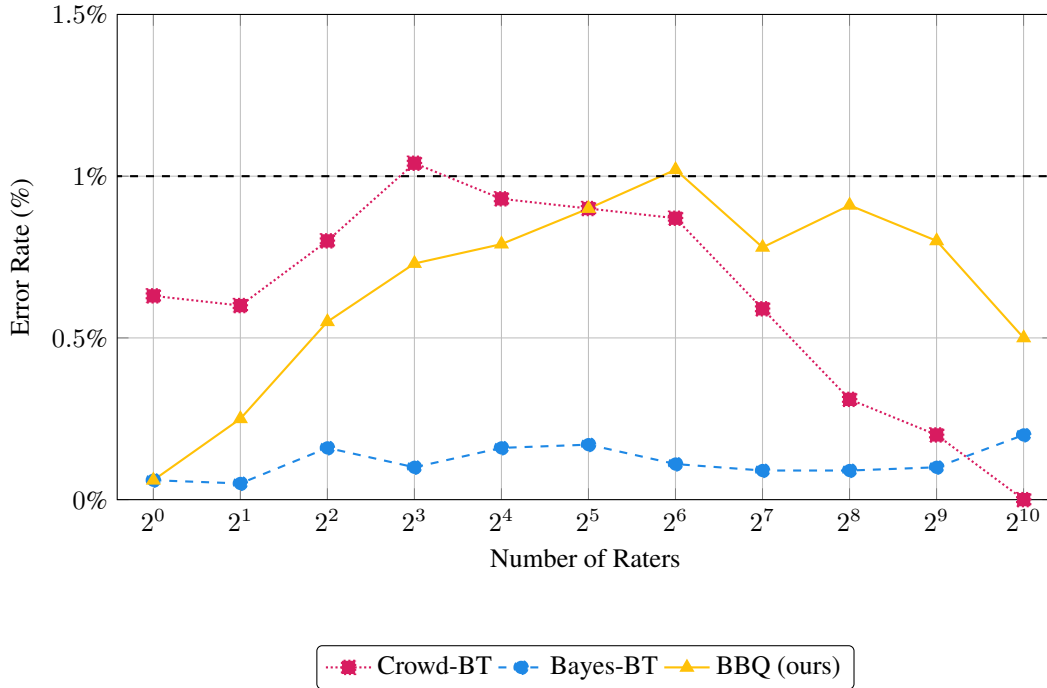


Figure 6: Type I error rate as the number of raters increases. The dashed line marks the expected 1% error rate.

To isolate rater quality, we split IHQ into screened and unscreened subsets. Screened raters passed pre-tests for color vision, display contrast, and sensitivity to subtle differences, and they completed training comparisons that reinforced the higher-quality-image criterion.

Figure 5 shows representative pre-screening images. The first two are standard color-vision tests requiring correct number identification, and the final two are low-contrast shape-detection tests using light-on-white and dark-on-black stimuli. These tests enforce a higher visual-quality threshold for the screened subset.

## F BOOTSTRAPPING DETAILS

Because BBQ and Crowd-BT explicitly model rater quality, we bootstrap raters rather than individual comparisons. This preserves each rater’s comparison distribution and yields a fair stability assessment for rater-quality models. For all datasets we use 10,000 rater bootstrap resamples, except HUMAINE, where one Crowd-BT bootstrap iteration takes approximately 15 minutes as discussed in Section 4.4. For HUMAINE, we therefore use 1,000 resamples to keep computation feasible.

## G UNCERTAINTY ESTIMATION DETAILS

Reliable rankings also require uncertainty estimates that indicate whether observed item differences are statistically meaningful.

Bayesian Bradley-Terry (Bayes-BT) and BBQ quantify uncertainty through the posterior distribution over item skills. Each skill has a Gamma prior updated by the observed pairwise comparisons. We convert posterior credible intervals to the Elo scale so item-level uncertainty is directly comparable.

Crowd-BT is non-Bayesian, so we estimate its uncertainty with a frequentist approximation. We use a second-order Taylor expansion around the maximum likelihood estimate. Inverting the Hessian of the log-likelihood yields a covariance matrix, whose diagonal entries give item variances. We convert these variances to 99% confidence intervals via:

$$p_{99} = \sqrt{\text{diag}(\text{covariance})} \times k_{\text{Erfc}0.01} \times \sqrt{2} \approx \sqrt{\text{diag}(\text{covariance})} \times 3.29,$$

where the constants scale the standard deviation to the 99% confidence level. Sharp likelihood peaks produce smaller intervals and higher certainty, while flatter peaks produce larger intervals and greater uncertainty.

We evaluate uncertainty calibration with a controlled simulation. Each trial contains two equally skilled items with 50–50 win probability, and pairwise outcomes are generated by coin flips. For every trial, we simulate multiple raters and apply each model to estimate Elo scores and uncertainty intervals.

Formally, the null hypothesis  $H_0$  states that the two items have equal strength. For each trial, we compute the 99% confidence or credible interval for each item’s skill estimate. A model incorrectly rejects  $H_0$  when the intervals do not overlap, implying a significant item difference even though none exists. The key metric is therefore the observed Type I error rate: the frequency with which a model falsely concludes that the items differ.

We run 10,000 trials with 50 comparisons per rater while varying the number of raters to measure how uncertainty scales with data. This setup estimates each model’s empirical Type I error rate and compares it with the theoretical 1% target at the 99% confidence level. As Figure 6 shows, Crowd-BT and BBQ stay close to the expected 1% error rate, demonstrating well-calibrated uncertainty estimates. Bayes-BT is consistently more conservative, producing lower-than-expected error rates and less suitable credible intervals for significance testing.